EdgeAnalytics_Observability_V1

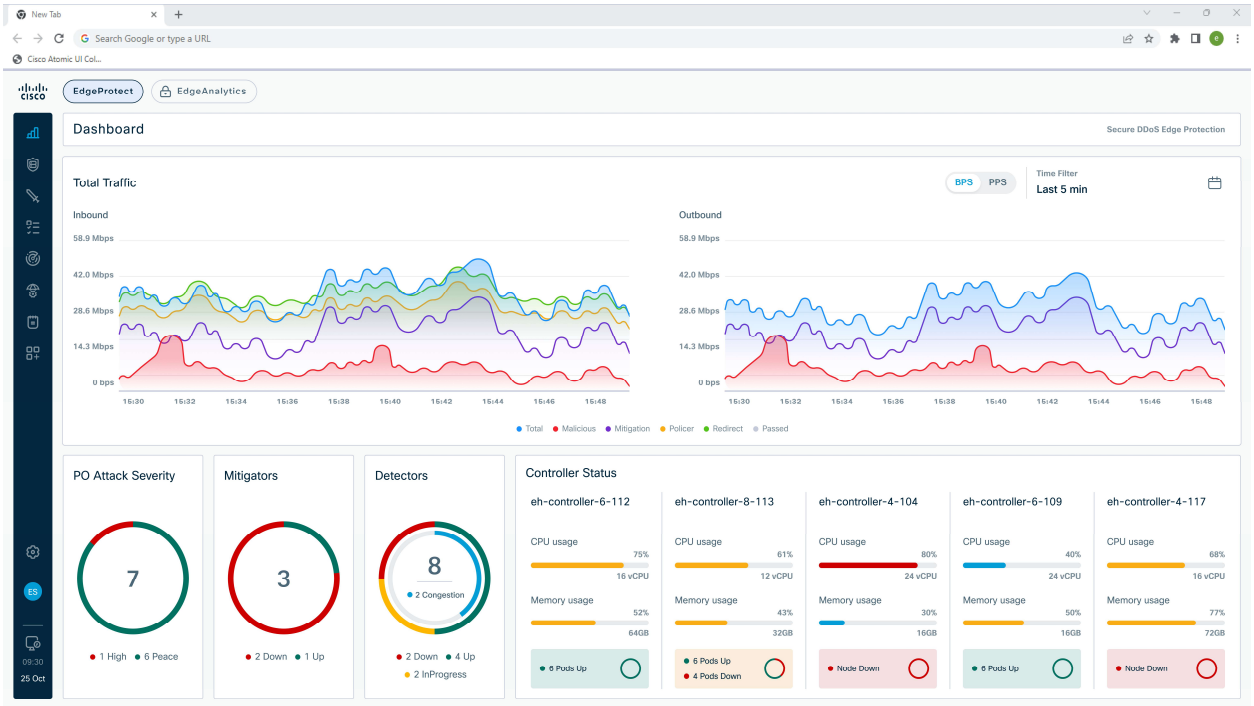
1 Overview

Our users are “requesting” and Observability platform to bridge the gap with companies like Arbor. In this platform EdgeProtect users will have access to EdgeAnalytics platform as an additional service. The objective of the Observability section within EdgeAnalytics is to allow the user to monitor their network’s traffic distribution and health for traffic monitoring and engineering.

The following user story will describe section by section how the first release should behave V1 or MVP.

2 Upgrading to EdgeAnalytics

- DDos EdgeProtect users (Administrators and clients with all of the rights such as Super Operator, Operators and Security Analysts) that have not purchased EdgeAnalytics liscence should see EdgeAnalytics tab in their tool.
- They will see the “button” with a small lock and greyed out:

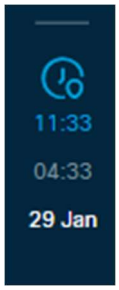


- When clicking on the button they will be prompted with a Pop-up prompting the user to send a customer support message similar to the one for license upgrade.
- Users that are EdgeAnalytics customers:
  - Here, the users with full rights would be able to switch between the two tools just by clicking on the top button tool
  - This user story will provide all of the details on how EdgeAnalytics will work for these type of users.

3 Fixed Areas and Contents:

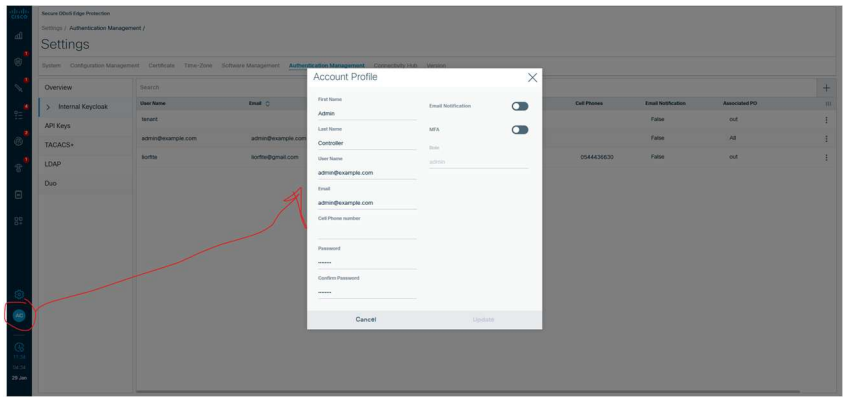
3.1 Clock Area

- Will be fixed among tools EdgeProtect and EdgeAnalytics



3.2 Account profile:

- For now, we will share the account profile screen
- **Admin:** Admins are the users that from EdgeProtect's Settings will be able to add and remove users to EdgeAnalytics
    - For V1 Admins can also add Super Operator, Operator and Security Analyst will have access to EdgeAnalytics (if subscribed to it)
    - Tenant users will not have access to it for V1
    - Non-admins that should not have access to EdgeAnalytics will see EdgeProtect as if EdgeProtect was the only tool
- UI will add some elements to: Settings>Authentication Management>Internal Keycloak> either new user or edit user
    - Toggle with EdgeAnalytics on/off. If on, then there will be a selection with checkboxes of what services the user can have access to:
        - ASN Path Analysis
        - Queries
            - Public queries
            - Private queries



4 Network Observability

4.1 Intro

Network Observability is the area that allows networks operators to understand the flows of traffic between an Origin and a Destination and all the points in between that affect their ASNs.

4.2 Icon

- Network observability will have its own icon and tooltip on the left panel
- For this MVP version the left panel will come only with one icon (Network Observability). And in subsequent version we will add more icons (Network Assurance, etc).

4.3 Data management

4.3.1 Data Collection

- Data Needed:
  - ASN IDs, Names, Owner and Geolocations (city and Country):
    - origin_asn
    - destination_asn
    - source_ip
    - destination_ip
    - Previous_asn
    - Next_asn
    - My Asns
  - My Routers info:
    - ASN (or ASNs if shared) that the router belongs to
    - Router name
    - Router ID
    - Router Location
    - Ingress interface name
    - Ingress interface ID
    - Egress interface name
    - Egress interface ID

    Note: One ASN could have many routers and each router many interfaces, also different ASNs can share the same router
  - Flow information:
    - Source
    - Destination
    - Protocol
    - Traffic in Gbps:
      - (Both for ingress and egress)
    - Packet count
    - Timestamp: Flow start time
- We will collect Netflow Data from the Client's network through a Netflow Collector and then save all of the info in a Time-Series Database
  - We will need to enrich and match the Netflow data with ASN names and GEOs from a DB

    - We will also enrich the data with DNS names
- DataLake → Clickhouse
- Data is collected directly from detectors or from the controller?
- The system should keep enriched NetFlow information in high enough resolution that will be sufficient to display data at a minimum time filter of 1 hour
- The storage (datalake?) holding all the information should be selected and designed in such a way that queries and presentations of data should be very fast (a 2 seconds max) and also once the data is loaded in a graph, "dive-in" into the graph should be instant
- **Sampling: needs to be defined**

4.3.2 Data Architecture

- As a user I would like to keep EdgeAnlaytics **separate and independent** from the EdgeProtect Controller
- As a user I would like to keep our database independent from both EdgeAnalytics and EdgeProtect?
- As a PM I would like to sell both products separately
- As a PM I would like to do MSSP on EdgeAnalytics ?? (not sure for V1)

4.3.3 Data retention

- The system shall be able to retain data for 1 year
- Once data is old or storage is about to be full, the data cleanup is FIFO

4.4 Snapshots & Backups

4.4.1 Snapshots

- I would like the user to be able to do snapshots to be able to restore EdgeAnalytics easily without the need to code all of the scripts again
- Snapshots should include everything:
    - Dashboard created by the user, ASN path analysis configuration, Queries created by the user
- **UI:** Inside Configuration management we should change it to 2 tabs: EdgeProtect and EdgeAnalytics
    - EdgeAnalytics tab will work the same way as EdgeProtect one:
        - First screen same table as in EdgeProtect
        - Icons- the same
        - Fields in the pop-up:
            - Observability
                - ASN Path Analysis
                - Queries
                - Dashboards
            - System Configuration
            - Authorization
            - System Integrations
            - Users
        - Actions on the side of each item: replace/append

4.4.2 Backups

- Databased should be backed up periodically
- Questions to answer?
  - Frequency: who defines it
  - Location: where are the backups saved
  - Ownership: do we do this from our system or does the user do it by himself
  - Retention: FIFO
  - Do we need UI for DB?

4.4.3 Upgrades

- Inside Settings > Software management > configuration management. We should add another tab under Controller and Detectors:
  - Name of the tab: EdgeAnalytics
    - Here Upgrades are performed as in the Controller, by pasting a link from Cisco
    - Revert upgrade process should be the same
    - UI similar as in EdgeProtect

4.5 ASN Path Analysis tab (V1)

4.5.1 General selectors

These are items that produce a general behavior in the ASN analysis tab.

4.5.1.1 Reporting

There will be a general reporting from the full ASN path analysis tab that could be downloaded by the user. Specific reporting of only pictures or only tables done

Reporting format:

- PDF
- HTML

4.5.2 ASN Path Analysis Tab

ASN path analysis tab will have several components described in sub-points of 4.4:

- Sankey diagram
  - My ASN Popup
  - Flow pop-up
- ASN Path analysis table

4.5.2.1 Sankey Diagram

This graph will be a **Sankey diagram** that will show the user the network flows in one view.

Everything described here ONLY affects the Sankey diagram

The Sankey diagram will have two main components:

- **ASNs:** which are visualized as rectangles
- **Flows**: Visualized as thick lines between rectangles

4.5.2.2 Filter

- The user can decide for the graph to filter based on the following "Stages" (only showing up in the following order):
    - **Origin ASN**
    - **Previous Peer**
    - **My ASN**
    - **Next Peer**
    - **Destination ASN**
- The minimum selection that can be made is 2 stages and the maximum is all of them
- Stages will always be in the order shown above, in the sense that for example the user selects only "Origin ASN" And "Next Peer" they will show up in that order
- Each "stage" is represented as a "vertical" in the ASN Path analysis graph
- Each Stage will have an associated color to them
- All of them once selected will show inside a **dropdown menu** with the option to search for a particular ASN inside that "stage" via either
    - ASN Name
    - ASN id
- The user can select ALL or deselect some or select only a few ASNs

4.5.2.3 Reporting

A picture of the ASN Path analysis would be able to be downloaded while clicking on a "photo" icon. Format: PNG. **(ALL graphs in EdgeAnalytics should have this PNG download icon)**

4.5.2.4 Behavior

- When the user passes the mouse on top of the graph tooltips will show up. There are different behaviors for different Areas of the graph.
- Passing the mouse on top of any **ASN tooltip:**
    - Header: Name of the ASN with ASN number
    - Traffic: Total number of GBs → this is the combination of GBs going IN or OUT the ASN
    - Country: Country from which
    - Number of In Flows: Number of Flows going in (The ASN that the user chooses as the first one will not have this)
    - Number of Out Flows: Number of Flows going out (The ASN that the user chooses as the last one will not have this)
- Passing the mouse on top of **flows tooltip:**
    - Header: Origin ASN name (number)-→Destination ASN name (number)
    - Traffic: Total number of GBs of the flow
    - Top Protocol: example TCP
    - Top Application: Port number and name (i.e: HTTPS: 443)
- Transformation of mouse "arrow" into a "hand": When the mouse transforms into a hand it means that is clickable. The following areas will be clickable (description of the pop-ups after this point)
    - My ASN rectangles → Opens a **My ASN Pop-up**
    - Flows → Opens **Flow Pop-up**

4.5.2.5 My ASN Pop-up

- When the user clicks on any ASN box a pop-up will open. This pop-up will close either by pressing "X" or by clicking outside the pop-up. All the screen that is outside the pop-up will be "semitransparent" and shaded, so the user knows intuitively than when he clicks out the pop-up closes.
- **The left side of the pop-up should be a tree topology table than when clicked the graphs on the right side should adapt to it**
- **This tree topology graph is persistent and it will keep its selection no matter what tab you choose. Changing the selection on the tree topology changes the contents of the tabs.**
- The right side of the graph will have several tabs
    - In these tabs the information shown would depend on what element is clicked on the table:
- **The table on the left side shows the contents in a tree** topology:
        - **My ASN name**
            - Router name
                - Interface name
    - Columns will be: **name – code or ID – Traffic in Gbps - % of Gbps**
    - **When click on any of these lines the graphs will change to adapt to it**
    - On the top right of the table there will be a calendar icon, allowing the user to change the time series that was previously selected in the Sankey screen without the need to go back to it
        - The change in time series if the user does it will affect the table and all of the tabs
    - Tree topology will have a search option
- Scheduler can be changed generally for all of the tabs if desired

4.5.2.5.1 Tab1: Routers & Interfaces tab

- **Sunburst chart.**
    - There are 2 sub-tabs for this chart: inbound and outbound
    - **Sunburst chart:**
        - When the user enters from a ASN he will see the Sunburst chart we 3 concentric circles:
            - Center Circle will show the following data:
                - My ASN name
                - Gbps
            - 1st concentric circle: Routers
                - Router name
                - % of Gbps that that specific router handles
            - 2nd concentric circle: Interfaces
                - Interface name
                - % of Gbps that that specific interface handles

○ The user will have the option to click (so the mouse should be shown as a hand) on the routers. When clicking an animation will show and the graph will change to:

- Center Circle:
  - Router name
  - Gbps
- Concentric circle:
  - Interface name
  - % of Gbps that that specific interface handles

○ When clicking back in the router (center) the graph will go back to full sunburst view

- **Hoovering over sunburst:**

  ○ When hoovering over the sunburst chart the specific legend row that the user is in will light-up/highlight – with our blueish brand tint

  ○ And it will allow the user to click if it's a router
- Notes:

  ○ If you click on an interface the user will just see a full circle with its contents

4.5.2.5.2 Tab 2: Traffic

- This will be a **time series staked**
- Legend is responsive
- Toggle: Gbps/pps
  - Graph Logic:
    - If the user has clicked or is by default in the ASN line:
      - This graph will show total ASN traffic with stacked lines that are routers traffic
    - If the user has clicked on a specific router:
      - This graph will show total Router Traffic with stacked interface traffic
    - If the user has clicked on an interface:
      - It will just show one line that it's the interface traffic
  - Is a stacked area graph with the following sections:
    - **Total** – the over all traffic In+Out+Transit+Internal
    - **Inbound** – Traffic arriving from ASN which are not MyASNs and terminating at MyASNs
    - **Outbound** – Traffic sent from MyASN to other ASNs which are not MyASNs
    - **Internal** - Traffic sent from MyASN to MyASNs
    - **Transit** - Traffic arriving from ASN which are not MyASNs and terminating at ANSs which are not MyASNs

4.5.2.5.3 Tab 3: Traffic per protocol

- **Bar chart /table view**
- Toggles & buttons:

        - Bar chart with/table view
        - Inbound/outbound:
            - By default it would be inbound
        - bps or pps:
            - By default it will be bps
        - Downloads:
            - All graphs will support direct PNG download from the graph
            - Tables will support CSV and Json
    - Tooltip on hoover for pie and line graph:
        - Traffic both in bps/pps and %
    - Graph:
        - Legend: TCP, UDP, ICMP, GRE, VPN, Others
    - Table view columns:
        - Protocol
        - %
        - bps
        - pps

4.5.2.5.4 Tab 4: Top 10 applications

- **Bar chart / table view**
- According to TCP/UDP port number as specified by IANA:
- Graph:
- Legend will be by name
    - Applications will be shown as:
        - HTTP /443
        - HTTP/80
        - DNS/53
        - BitTorrent/6881
        - SSH/22
    - Tooltip on hoover on the bar
        - Will show the name (same one as in the legend)
        - Traffic both in bps/pps
        - Percentage in %
- Table view columns:
    - Name
    - Application
    - Port number
    - %
    - bps
    - pps

4.5.2.6 Flow Pop-up

- This pop-up will show once the user clicks on a specific flow on the Sankey diagram
- Flow name: ASN (ID) → ASN(ID)

4.5.2.6.1 Graph 1: Traffic per Protocol:

- **Bar chart /table view**
- Toggles & buttons:
    - Bar chart with/table view
    - Inbound/outbound:
        - By default it would be inbound
    - bps or pps:
        - By default it will be bps
    - Downloads:
        - All graphs will support direct PNG download from the graph
        - Tables will support CSV and Json
- Tooltip on hoover for pie and line graph:
    - Traffic both in bps/pps and %
- Graph:
    - Legend: TCP, UDP, ICMP, GRE, VPN, Others
- Table view columns:
    - Protocol
    - %
    - bps
    - pps

4.5.2.6.2 Graph 2: Top 10 applications

- **Bar chart / table view**
- According to TCP/UDP port number as specified by IANA:
- Graph:
    - Legend will be by name
        - Applications will be shown as:
            - HTTP /443
            - HTTP/80
            - DNS/53
            - BitTorrent/6881
        - SSH/22
    - Tooltip on hoover on the bar
        - Will show the name (same one as in the legend)
        - Traffic both in bps/pps
        - Percentage in %
- Table view columns:
    - Name
    - Application
    - Port number
    - %
    - bps
    - pps

4.5.2.7 ASN Path Analysis Table

This table will contain all the aggregated information **for the flows** showing in the Sankey diagram.

4.5.2.7.1 Reporting

Table will be available to be downloaded in:

- CSV
- Json

4.5.2.7.2 Columns

The columns shown will be based on the filters selected in the General Selectors of the path analysis page.

Column name	Content
Origin	Name/Number
My ASN	Name/Number
Destination	Name/Number
Origin Geo	city/country
Destination	city/country
Traffic	Gbps (average traffic speed)
Total traffic	TB for the period
Flow count	Number of flows for the period
Average flow size	MB
Peak traffic	Gbps
Protocol mix	60% TCP, 40%UDP, xx%, other%
Top port	443 (HTTPS) 65%
Trend traffic speed	Increase of 15% vs previous period* from the average traffic in Gbps
% of Total	12.5% (portion of the traffic going through My ASN) from the average traffic in Gbps

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4.6 Queries Tab (V1)

Queries will allow the user to create his own graphs by querying into our database.

4.6.1 Downloadable content:

- Images as PNG
- Tables as CSV/Json
- Full Dashboards as PDF or HTML

4.6.2 General

Queries will have some common UI components between them:

- Time: the time selector will be common in an query/dashboard page
- Queries will have 2 tabs:
  - Query dashboard

    - My Queries

4.6.3 Query dashboard

- Here the user can add the graphs and or tables of the queries he has done himself
- He should be able by drag and drop to reorganize them and scale them as he wishes → essentially we should be able to provide a responsive and customizable screen so the user can create his own dashboards
- Saving dashboards:
    - Query dashboard will allow the users to save a particular dashboard
    - For doing this there will be a "save" option that will allow the user to give a name to the dashboard. For example, a user might want to call a particular dashboard TCP view, because he might want to see different graphs and flows of data that focuses on TCP protocol
- Needless to say that a dashboard is composed by a set of graphs, and each graph is developed in a query.
- You should be able to add new graphs/tables by clicking on a "+".
    - After clicking the user will open a pop-up that will open a pop-up showing the list of "my queries" saved (and NOT the draft ones).
    - This pop-up will show the table with the available queries on the left and when clicking on them a preview of the query will be shown on the right.
    - Each column will have a checkbox.
    - The user can select one or multiple queries to Add
    - The bottom of the screen will have two buttons: "Cancel" and "Add"
    - The user can click outside the pop-up and it will work as "Cancel"
    - Here the user will select an existing query or create a new one.
    - As V1 we will only show the queries saved in my queries and not "draft" or "create new queries" from this point
- Users should be able to do the following actions on Each graph/table:
    - Download as PNG (if graph) or CSV/Json (if table)
    - Edit: in which case a pop-up with the query will open in which the user will be able to edit the query. Behavior described in "My queries".
        - After saving the edit, the edit is automatically saved in "my queries" too.
    - Delete graph/table from the dashboard: for the avoidance of doubt, this does NOT delete the query from "My Queries".
- After any change has been made to any component of the dashboard (including the general scheduler) the user will be able to save the changes.
- User should also be able to change the name of the dashboard
- When the user creates a dashboard from the first time, when clicking "save" the user will be prompted with a pop-up that asks him to give a name to the dashboard:
    - This name should be a short identificatory name
    - He can also add a Description – that we will show as a tooltip when the user hoovers over the dashboard name

- After the user has saved his first dashboard then a select dashboard drop down will be available where the dashboard name is, here the user can select and change the dashboard between all of the available dashboards that he created.
- When we are looking at a dashboard created we should also have a button which is: New Dashboard to create a new dashboard. Here we enter the dashboard creation flow.

4.6.4 My queries

In this section we describe how does the user generates and stores his queries and manages them. We will also talk about how the script should look like and what actions it should perform, and the look and feel of this system.

- This section has the following components:
    - On the top: There is a "+" to add a new query.
    - The main screen is partitioned in 2:
        - Left side: Will have 2 tabs Queries/Drafts. Both of them will behave the same way, the only difference is that Drafts queries will NOT be available to be chosen in the dashboard.
            - List of the queries the user has saved shown as a table with the following columns/components:
                - Name (with filter and sort)
                - Description
                - Last edited (with sort)
                - Each row will have a 3-dot menu to allow the user to:
                    - Edit – it will open the query edit pop-up
                    - Clone – it will open a pop-up asking you to give the query a name. With three buttons, cancel , edit and save. If saved the query goes to the query list, if edit it opens the query editor
                    - Delete. It will open a pop-up asking. Are you sure you want to delete this query? Buttons Cancel/Yes
                - Double click in any row will open the editing menu of the query
        - Right side: preview of the query as an image/table depending on what the query is
            - There will always be on top a "calendar" to select the time-default 1h

4.6.4.1 Writing a query

- The query screen that will open will be a pop-up and it will be partitioned:
    - Left side will be the **Query section:**
        - Top section:
            - Area to enter a title (grey faded text "Write your title" – to indicate that the user can write the query title)
            - The query to be viewed on the right side: auto preview as the user types valid queries

        - Here the user will write its query, which operators and functionality is described in Query operators
    - Right side will be a preview of the query as a graph/table depending on what operators is the user using:
        - Note: the graph/table will always have on top a time series calendar by default in 1h, but the user can change it
        - The graph/table will show its title on the top
    - At the bottom of the screen there will be 3 buttons "cancel", "save as draft" or "save".
        - Save as draft saves the query in the draft section, which means it will not be available to be chosen in the dashboard
        - There will not be "x" on the top of the screen and
        - clicking outside the pop-up will not close it, but rather it will save it automatically as draft

4.6.4.2 Query operators

- Important:
    - the system is automatically debugging and checking the validity of the operators and/or wrong input
    - same behavior as in mitigators: "space" will be for autocomplete, and "tab" for accepting the selection within the auto-complete
- Condition selects one of the following and always **equals** it to a value:
    - **,** → we use commas for a list separators **everywhere**
    - Dimensions:
        - Src.Country
        - Dst.Country
        - Src.City
        - Dst.City
        - Src.ASN
        - Dst.ASN
        - Prev.ASN
        - Nxt.ASN
        - Src.PO
        - Dst.PO
        - Src.Port
        - Dst.Port (application)
        - Src.IP (CIDR)
        - Dst.IP (CIDR)
        - Protocol
        - Application
        - Router.name
        - Router.ID
        - Interface.name (on a Router)
        - Interface.ID (on a Router)
        - Direction.Inbound

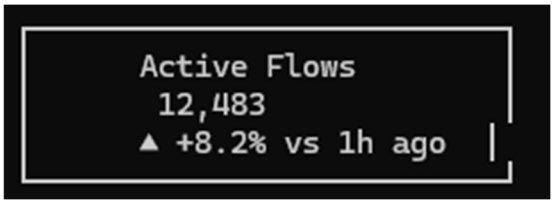
        -
        - Hour (shows the hour of the day)
        - Day.week (shows the day of the week as Monday, Tuesday, etc)
        - Month (shows the month)
        - Mitigation.rule: shows which rule the client is firing most: ACL, Flowspec, RTBH
    - Metric:
        - Traffic → bps
        - Packets → pps
        - Flows → Flow count
        - Bytes → total bytes
        - GB
        - TTL → Seconds

    - Conditions:
        - Logical operators:
            - AND
            - OR
            - AND NOT
            - OR NOT
        - Comparison operators for name values:
            - IS
            - IS NOT
        - Comparison operators for numeric values:
            - == “equal ”
            - != “not equal
            - > “bigger than”
            - >= “bigger or equal”
            - < “smaller than”
            - <= “smaller or equal”
            - BETWEEN n AND n : i.e: BETWEEN 50 AND 100
    - Arithmetic operations:
        - * Multiplication
        - / Division
        - - Subtraction
        - + Addition
    - Aggregation
        - AVERAGE it calculates the average of an array
        - MEDIAN it calculates the median of an array
        - TOTAL it sums all the array
        - COUNT it counts the amount of times a specific value appears
        - MIN – minimum value
        - MAX – maximum value

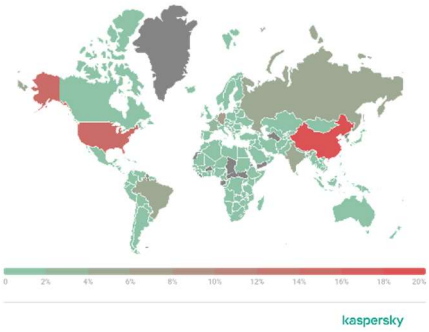
        - P95
        - P99
        - () → for doing operations and formulas
    - Graph types (descriptions and examples on how to use them in the script in following sections):
        - Line graphs:
            - Time Series**:**
                - Syntax: **TIME SERIES**
                - it will show a time series in which the horizontal axis is the time selected in the selector and the vertical axis is the result of the operators selected in the script.
                - Legend will be the names selected (i.e: routers)
                - Note: Several different data selected in time series will NOT stack them up but rather added at the value they have
            - Stacked Area:
                - Syntax: **STACKED AREA**
                - Like time series but shows composition over time.
                - Areas are stacked on top of each other
        - Circular charts:
            - Pie Chart**:**
                - Syntax: **PIE**
                - automatically for any time it will calculate % over the set of values of a period.
                - The values over which the Pie chart is calculating the % need to be defined by the user and they could be any arithmetic operation (example: TOTAL Protocol IS TCP)
                    - By default if no calculation is being made values will be the average of the period
                - The name of the legend will be an array of data selected by the user (example: Router name)
            - Gauge chart
                - Syntax: **GAUGE**
                -
                - A single big number with optional threshold coloring
            -

        - Single Value box**:**
            - Syntax: **SINGLE VALUE**
            - displays a single value with its title, i.e:

- Or if programmed accordingly a single value with a comparison of previous period

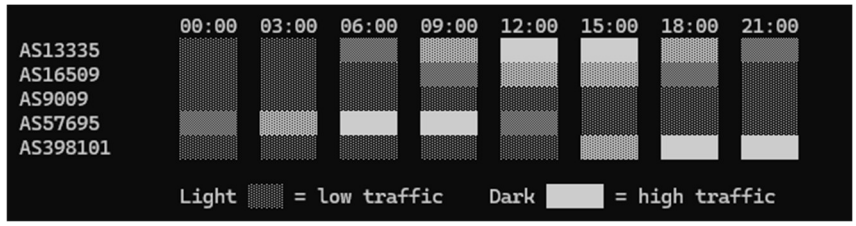


- Bar charts:
  - Vertical lines Bar chart:
    - Syntax: **BAR**
    - Horizontal axis will be the name of the data selected (example: Router name) and this will be the name of the legend
    - Vertical axis will be the result of a user operator
      - By default: if not calculations is being made, values will be the total of the period
  - Horizontal lines bar chart:
    - Syntax: **TOP-N**
      - i,e: TOP-5
      - i,e: TOP-10
    - Horizontal axis will be the units or the result of a user operator
    - Vertical axis will be the name of the data selected (i.e protocol)
    - Note: Cleaner than vertical bars when you have long labels (ASN names, IP addresses).
- Geography Map:
  - Syntax: **GEO**
  - To plot traffic on a world map with different colors as a heatmap

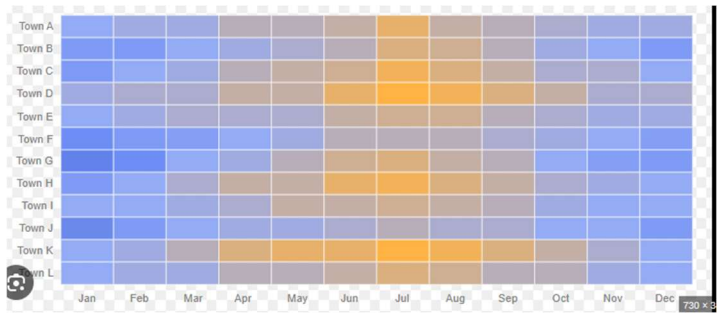


  - Reference:
- Heatmap chart:

- Syntax: **HEATMAP**
- Provides a heatmap of values on columns X and Y
- Reference of what it can be useful for → identify if there is more traffic per hour at a specific ASN



- Visual reference:



- Sankey graph:
  - Syntax: **SANKEY**
  - Creates a Sankey graph
- Table graph:
  - Syntax: **TABLE**
  - creates a table with rows and columns

4.6.4.3 Query base structure

- The following query base structure applies to all queries

[**LET** **<name> = <expression>**] → *Allows the user to declare a reusable variable that as default is the axis label → for formulas, i.e: LET throughput_gbps = SUM(bytes) * 8 / 1000000000 → variable name becomes default axis label → "throughput_gbps"*

[**/* comment */**]

<GraphType>

<Aggregation> <Metric> [**AS** **<"display label">**] → *Overrides the display label, for declarations use LET. i.e if we continue with the previous example and we write: AS "Throughput" → overrides the display label → "Throughput"*

[**BY** **<Dimension>**[, <Dimension>]]

[**WHERE** **<Conditions>** [AND|OR <condition>]]

*NOTES:

- Anything in between [] is optional
- Order of the Query needs to be as described above → order matters

4.6.4.3.1 <GraphType> modifiers

Each graph type accepts specific additional keywords. These keywords go after **WHERE** at the end of the query

- Modifier: **BY**
  - Purpose: Group by <Dimension>
    - TIME SERIES: 1 dim
    - STACKED AREA: 1 dim
    - PIE: 1 dim
    - GAUGE: — → GAUGE never uses BY as it produces one value.
    - SINGLE VALUE: —→ SINGLE VALUE never uses BY as it produces one value.
    - BAR: 1 dim
    - TOP-N BAR: 1 dim
    - GEO MAP: 1 geo dim → GEO MAP BY must be a geographic field (country, src_country, region, etc).
    - HEATMAP: 2 dims → HEATMAP requires exactly 2 dimensions in BY (rows × columns).
    - SANKEY: — → SANKEY uses FROM/TO instead of BY (see below)
    - TABLE: 1+ dims

- Modifier: **FROM / TO**
  - Purpose: Flow source → target
    - TIME SERIES: —
    - STACKED AREA: —
    - PIE: —
    - GAUGE: —
    - SINGLE VALUE: —
    - BAR: —
    - TOP-N BAR: —
    - GEO MAP: —
    - HEATMAP: —
    - SANKEY: required → SANKEY uses FROM/TO instead of BY.
    - TABLE: —

- Modifier: WHERE
  - Purpose: Filter data
    - TIME SERIES: yes
    - STACKED AREA: yes
    - PIE: yes
    - GAUGE: yes
    - SINGLE VALUE: yes
    - BAR: yes

        - TOP-N BAR: yes
        - GEO MAP: yes
        - HEATMAP: yes
        - SANKEY: yes
        - TABLE: yes

- Modifier: LIMIT <n>
    - Purpose: Cap results
        - TIME SERIES: yes → LIMIT on TIME SERIES caps how many series lines/areas are shown (keeps top N by value, avoids 100 overlapping lines).
        - STACKED AREA: yes → LIMIT on STACKED AREA caps how many series lines/areas are shown (keeps top N by value, avoids 100 overlapping lines).
        - PIE: yes
        - GAUGE: —
        - SINGLE VALUE: —
        - BAR: yes
        - TOP-N BAR: requires a limit→the N is the LIMIT. Without it, it's just a BAR.i.e: TOP-5
        - GEO MAP: —
        - HEATMAP: —
        - SANKEY: —
        - TABLE: yes

- Modifier: **COMPARE PREVIOUS** <period>
    - Purpose: Delta vs past period
        - TIME SERIES: —
        - STACKED AREA: —
        - PIE: —
        - GAUGE: —
        - SINGLE VALUE: yes
        - BAR: —
        - TOP-N BAR: —
        - GEO MAP: —
        - HEATMAP: —
        - SANKEY: —
        - TABLE: —

- Modifier: MIN <n> MAX <n>
    - Purpose: Gauge range
        - TIME SERIES: —
        - STACKED AREA: —
        - PIE: —
        - GAUGE: required

- SINGLE VALUE: —
- BAR: —
- TOP-N BAR: —
- GEO MAP: —
- HEATMAP: —
- SANKEY: —
- TABLE: —

- Modifier: THRESHOLDS <n>, <n>
  - Purpose: Green/yellow/red zones
  - TIME SERIES: —
  - STACKED AREA: —
  - PIE: —
  - GAUGE: required
  - SINGLE VALUE: —
  - BAR: —
  - TOP-N BAR: —
  - GEO MAP: —
  - HEATMAP: —
  - SANKEY: —
  - TABLE: —

4.6.4.4 Query Examples

```
/* Example 1: Time series — Inbound traffic per router */
TIME SERIES TOTAL(Traffic)
  BY Router name
  WHERE Dst.PO IS "Madrid"
```

Result: Line chart, X-axis = time, one line per router, Y-axis = total traffic.

```
/* Example 2: Stacked area — Protocol composition over time */
STACKED AREA TOTAL(Traffic)
  BY Protocol
  WHERE Router name IS "core-router-01"
  LIMIT 5
```

Result: Stacked area chart showing top 5 protocols stacked over time.

```
/* Example 3: Bar chart — Top routers by traffic */
```

```
BAR TOTAL(Traffic)
  BY Router name
  WHERE Src.Country IS "US", "DE", "JP"
```

Result: Vertical bars, one per router, filtered to traffic from US, Germany, Japan.

```
/* Example 4: Top-N horizontal bar — Heaviest ASNs */
TOP 10 BAR TOTAL(Traffic)
  BY Src.ASN
  WHERE Protocol IS TCP
```

Result: Horizontal bar chart, top 10 source ASNs ranked by TCP traffic.

```
/* Example 5: Pie chart — Protocol distribution */
PIE TOTAL(Traffic)
  BY Protocol
  WHERE Router name IS "edge-router-02"
  LIMIT 6
```

Result: Pie chart showing % share of each protocol on that router.

```
/* Example 6: Single value — Total current traffic */
LET total_gbps = TOTAL(Bytes) * 8 / 1000000000

SINGLE VALUE total_gbps AS "Total Inbound (Gbps)"
  WHERE direction IS "inbound"
```

Result: Big number showing e.g. "847" with label "Total Inbound (Gbps)".

```
/* Example 7: Single value with comparison */
SINGLE VALUE COUNT(Flows) AS "Active Flows"
  WHERE status IS "active"
  COMPARE PREVIOUS 1h
```

Result: Big number "12,483" with delta "▲ +8.2% vs 1h ago".

```
/* Example 8: Gauge — Controller CPU */
GAUGE AVERAGE(cpu_usage) AS "Controller CPU (%)"
  WHERE Router name IS "eh-controller-8-112"
```

```
  MIN 0 MAX 100
  THRESHOLDS 60, 85
```

Result: Speedometer dial at 79%, yellow zone.

```
/* Example 9: Heatmap — Attack patterns by ASN and hour */
HEATMAP COUNT(Flows)
  BY Src.ASN, Hour
  WHERE Protocol IS UDP
```

Result: Grid, rows = ASNs, columns = hours, color = flow count intensity.

```
/* Example 10: Sankey — Traffic flow between source and destination ASNs */
SANKEY TOTAL(Traffic)
  FROM Src.ASN TO Dst.ASN
  WHERE Src.Country IS "BR"
```

Result: Sankey flow diagram showing how Brazilian traffic routes between ASNs.

```
/* Example 11: Geo map — Attack traffic by country */
GEO MAP TOTAL(Traffic)
  BY Src.Country
  WHERE Mitigation rule IS "ACL", "Flowspec"
```

Result: World map heatmap, countries colored by volume of mitigated traffic.

```
/* Example 12: Table — Detailed flow analysis */
TABLE TOTAL(Traffic), COUNT(Flows), AVERAGE(TTL)
  BY Src.ASN, Dst.Port, Protocol
  WHERE Src.Country IS "CN"
  LIMIT 50
```

Result: Table with 50 rows, columns: Src.ASN, Dst.Port, Protocol, Total Traffic, Flow Count, Avg TTL.

```
/* Example 13: LET with formula — Bits per packet ratio */
LET bpp = TOTAL(Bytes) * 8 / TOTAL(Packets)
TIME SERIES bpp AS "Bits per Packet"
  BY Src.ASN
```

```
WHERE Protocol IS TCP
LIMIT 5
```

Result: Time series showing bits-per-packet ratio per ASN — useful for detecting small-packet DDoS floods.

4.6.4.5 Not mandatory: LLM – describe a query via natural language

- Ideally, at the button of the script editor, there will be a chat input in which the user can write in natural language what graph he wants to see and the query will auto-write.